

Problem Statement: Traditional filters fail to detect modern **AI-generated phishing emails**. There is no single platform that can detect threats **without data leaks**, **trace attackers behind VPNs/proxies**, and generate **instant forensic evidence** for quick response.

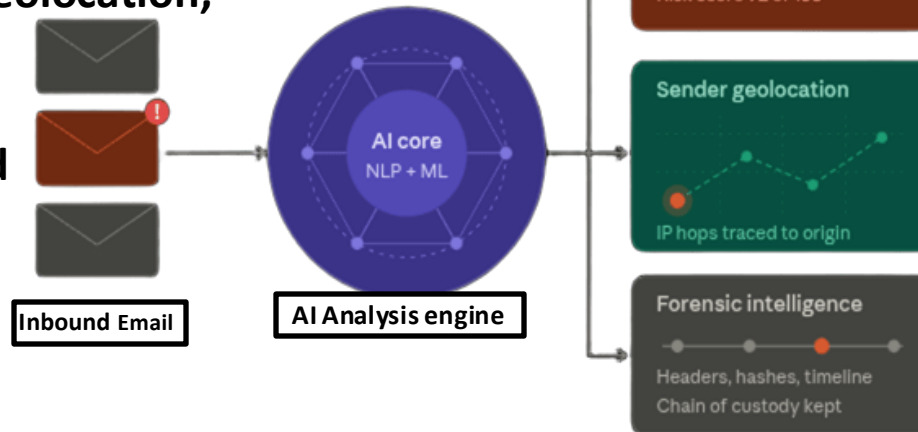
Our Idea: An AI platform where you upload an email, **it detects phishing/spoofing**, traces the attacker's real location behind VPN/Proxy, and instantly **creates a forensic report** for quick action.

Proposed Solution:

- It **detects AI-generated phishing**, compromised-account abuse, impersonation, and **BEC using multi-layer detection**(content analysis, sender authentication, behavioral intelligence, **infrastructure geolocation**, and **threat correlation**).

- They show **why an email was flagged** and **how the incident evolved**, with related campaigns clustered together.

- analyst feedback cuts false positives, **quarantining** and **blocking** contain threats, and **forensic-grade evidence supports** incident response.

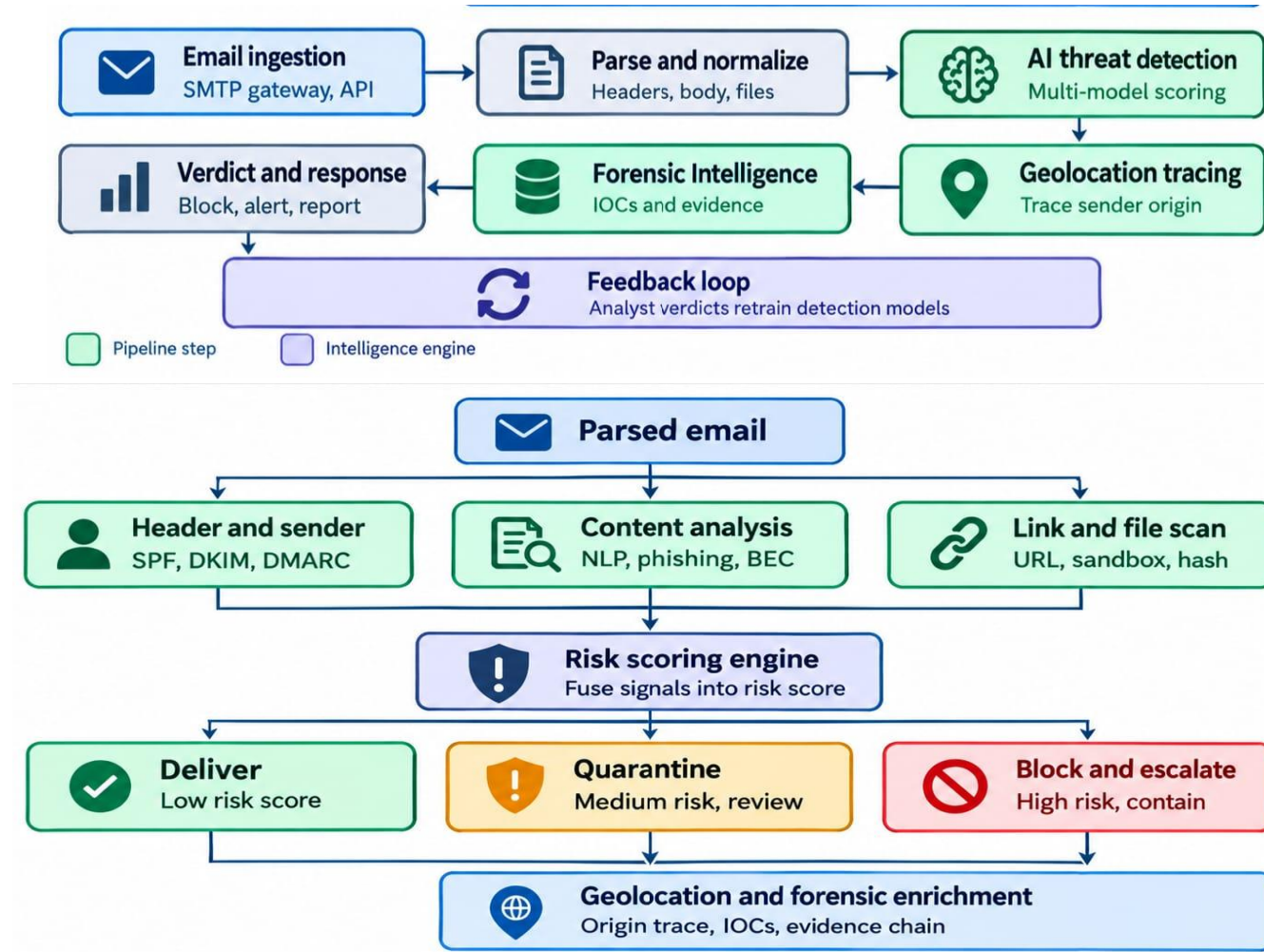


Innovation/Uniqueness:

- Autonomous, explainable, and privacy-preserving agent:** it acts on its own, justifies every decision, and protects user data.
- Behavioral identity analysis:** it learns normal sender and user behavior to spot impersonation and compromised accounts.
- AI-generated phishing detection:** it catches machine-written phishing and BEC content that traditional filters miss.
- Infrastructure geolocation and threat-intelligence graphs:** it traces here attacks originate and links senders, domains, and IPs into a connected view

Hardware And Software

- **Architecture and email ingestion:** the overall platform design, plus how emails are collected and normalized for analysis.
- **Threat detection engines:** URL and attachment analysis, along with identity, authentication, and relationship analysis.
- **Geolocation and infrastructure intelligence:** tracing where senders, domains, and IPs originate and how they connect.
- **Forensic intelligence, case management, and response:** evidence-grade investigation and reporting, plus automated orchestration such as quarantine and blocking.
- **Continuous improvement and governance:** model training, evaluation, and feedback loops, together with security, privacy, and compliance controls.



Feasibility

- **AI/ML detection:** NLP models catch phishing and BEC Accurately
- **Header forensics:** SPF/DKIM/DMARC and Received header parsing is mature.
- **IP intelligence:** Threat feeds and geolocation enable origin and risk scoring.
- **Graph correlation:** Links senders, domains, and IPs to expose campaigns.
- **Easy integration:** M365, Google Workspace, and SIEM/SOAR tools (Splunk, Sentinel) connect via standard

Commercial Feasibility

Growing market: Demand for AI-based email security is increasing rapidly.

- **Strong need:** Organizations need protection from phishing, fraud, ransomware, and spoofing.
- **Unique solution:** Combines threat detection, geolocation, and digital forensics in one platform.
- **Revenue potential:** Can earn through SaaS subscriptions and premium enterprise services.

Challenges

- **Accuracy risks:** False positives and missed attacks can reduce user trust.
- **Evolving threats:** Attackers constantly change tactics and try to evade AI detection.
- **Unreliable evidence:** Geolocation and email headers may be incomplete, misleading, or inaccurate.
- **Privacy and legal risks:** Data collection requires consent, regulatory compliance, and secure handling.

Strategy

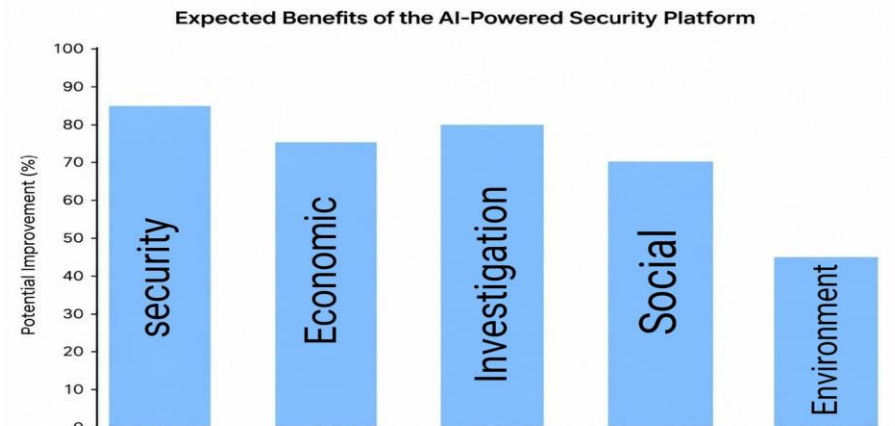
- **Detect threats with AI:** Check emails, links, attachments, sender identity, language, and authentication for phishing and malware.
- **Analyze location and identity:** Detect unusual IP locations, impossible travel, VPNs, fake domains, and suspicious login behavior.
- **Respond automatically :** Score threats, quarantine dangerous emails, block links, and alert security teams quickly.
- **Preserve forensic evidence:** Store email headers, timestamps, IPs, URLs, files, user activity, and investigation records.

Potential Impact on the targeted users

- **Faster threat detection:** Helps security teams quickly identify phishing, spoofing, scams, and malicious emails.
- **Better cybercrime investigations:** Helps government and police departments trace email sources, locations, and criminal activity.
- **Reduced business losses:** Protects companies and banks from fraud, data theft, financial loss, and reputational damage.
- **Easy security for institutions:** Gives schools, colleges, and public offices simple risk alerts, even with limited cybersecurity staff.
- **Safer email use:** Warns employees and customers about suspicious links, fake senders, unusual locations, and other signs of fraud

Benifits of solution

- **Improved security:** Detects phishing, malware, spoofing, and other email threats earlier, reducing cyberattacks and data breaches.
- **Economic savings:** Minimizes financial losses, fraud, downtime, and the cost of manual investigation and incident response.
- **Faster investigations:** Uses geolocation and forensic intelligence to quickly identify attack sources, affected users, and evidence.
- **Social protection:** Safeguards individuals, businesses, and public institutions from identity theft, scams, privacy violations, and misinformation.
- **Environmental benefit:** Reduces paper-based reporting, unnecessary travel, and energy-intensive manual investigations through automated digital analysis.



AI/ML Phishing and Email Threat Detection

- [Deep Learning-Based IDS for Phishing Email Detection: A Short Survey \(ICCV Workshops 2025\)](#)
Reviews CNN, RNN, Transformer and hybrid or multi-modal models, along with datasets, adversarial manipulation, and combining URL and header cues.
- [Enhancing Phishing Email Identification with LLMs \(arXiv 2502.04759\)](#)
Covers LLM-based phishing email classification.

Datasets

- [Champa et al., ISDFS 2024](#)
Describes curated versions of these datasets and their class imbalance
- [IEEE DataPort: Phishing Email Detection Datasets](#)
Includes a combined dataset with extracted features such as URL statistics, attachment indicators and SPF results.

Email Forensics and Header Analysis

- [Tracing the Geographic Origin of an Email](#)
Explains how to extract IPs from Received headers and map them to locations.
- [Reading and Interpreting Received Headers](#)
Gives a step-by-step workflow, including the SPF, DKIM and DMARC checks.

IP Geolocation

- [MaxMind: Geolocation Accuracy and Limitations](#)
States that the data is never precise enough to identify a specific household, individual or street address.
- [MaxMind GeoLite2 \(Free Databases\)](#)
Country, City and ASN databases updated twice weekly; free but less accurate than paid GeoIP2.

TEAM APEX INNOVATORS

**Sagar Vaishnav**Team Leader &
Project Coordinator**Nikhil Patidar**Research &
Data Analyst**Vaidik Purohit**Email Forensics &
Security Analyst**Darshna Jain**Backend &
Streaming Engineer**Vinayak**AI/ML &
Threat Detection Engineer**Jinal Verma**Dashboard &
Validation Engineer